Survival Analysis

Assignment 2

Non-Parametric, Semi-Parametric & Extended Cox Models

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Outline

1. INTRODUCTION
2. METHODOLOGY
3. RESULTS AND INTERPRETATION
4. CONCLUSION

INTRODUCTION

Objectives

General Objective

To obtain an assumption-robust characterisation of survival patterns and covariate effects.

Specific Objectives

1. To estimate survival empirically (Kaplan-Meier)
2. To Characterise cumulative hazard (Nelson-Aalen)
3. To test group differences with multiple weighted tests (Log-Rank, Breslow, Tarone-Ware, Fleming-Harrington)
4. To model covariate effects via Cox PH
5. To diagnose, refine, and extend the Cox model (log-log plots, time interactions)
6. To synthesise all findings

Research Questions

1. What does the Kaplan-Meier curve reveal about overall survival?
2. What shape does the cumulative hazard take (Nelson-Aalen)?
3. Are survival differences confirmed across all weighted tests?
4. Which covariates significantly affect the hazard (Cox PH)?
5. Do log-log plots confirm PH, and what does the spline model reveal?
6. Does the Lev+5FU treatment effect change over time?
7. How do all approaches converge?

METHODOLOGY

About the data

Dataset

The `colon` dataset (North Central Cancer Treatment Group) contains 929 patients from a randomised adjuvant chemotherapy trial. Key variables: treatment (`rx`: Obs, Lev, Lev+5FU), age, sex, nodes (positive lymph nodes), and tumour differentiation.

Table 1: Summary of Patient Data in the Colon Cancer Dataset

	n	**%**
Total patients	929	100.0
Events (deaths)	452	48.7
Censored	477	51.3

- The dataset contains 929 patients, with 452 events (deaths) and 477 censored observations.

Data Preparation

1. Restricted to **death events only** (`etype = 2`), one record per patient
2. **Missing data:** `nodes and differ` had $< 2\%$ missing values, imputed using Last Observation Carried Forward (LOCF)
3. **Factor encoding:** `sex` (Female/Male), `rx` (Obs/Lev/Lev+5FU), `differ` (Well/Moderate/Poor)
4. **Survival object:** constructed as `Surv(time, status)` in R

Hypothesis Testing

Log-Rank & Wilcoxon Tests

To formally test for survival differences between groups, two non-parametric tests were applied:

$$\chi^2 = \frac{\left(\sum_j w_j (O_{1j} - E_{1j})\right)^2}{\sum_j w_j^2 V_j}$$

where O_{1j} and E_{1j} are observed and expected deaths in group 1 at time t_j, and V_j is the hypergeometric variance.

- **Log-Rank** ($w_j = 1$): equal weight at all time points; optimal under proportional hazards; sensitive to late differences.
- **Wilcoxon/Breslow** ($w_j = n_j$): weights early events more heavily; more sensitive to early survival differences.

Cox Proportional Hazards Model

Cox PH Model

The Cox proportional hazards model relates covariates to the hazard without specifying the baseline hazard $h_0(t)$:

$$h(t \mid \mathbf{x}) = h_0(t)\,\exp\!\big(\beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p\big)$$

Parameters are estimated via **partial likelihood**. The **hazard ratio** for covariate x_k is $\mathrm{HR}_k = e^{\hat{\beta}_k}$, interpreted as the multiplicative change in hazard per unit increase in x_k.

Modelling Strategy

1. **Univariable models:** one covariate at a time: age, sex, rx, nodes, differ
2. **Multivariable model:** all covariates simultaneously, yielding adjusted hazard ratios

Diagnostics & Model Refinement

PH Assumption & Residual Diagnostics

Three diagnostic tools were applied to the multivariable model:

1. **Schoenfeld residuals** (`cox.zph`): test H_0: scaled residuals are uncorrelated with time. A significant p-value indicates PH violation.
2. **Martingale residuals**: plotted against continuous covariates (age, nodes) to detect non-linearity in the log-hazard. A non-flat LOESS trend indicates a violation of the linearity assumption.
3. **dfbeta residuals**: measure the influence of each observation on each coefficient. Values outside ± 0.02 flag influential patients.

■ **Refined model:** where non-linearity was detected (nodes), a **penalised spline** (`pspline(nodes, df=4)`) was incorporated to flexibly model the log-hazard relationship.

Synthesis & Software

Synthesis

Results from all methods were integrated to answer whether the non-parametric, semi-parametric, and parametric (GenGamma, Project 1) approaches converge in their characterisation of:

- The shape of the hazard over time
- The magnitude and significance of treatment effects
- The key predictors of survival

Software

All analyses were performed in **R** using the following packages:

- `survival`: KM, NA, Cox PH, diagnostics
- `survminer`: KM plots with risk tables
- `flexsurv`: GenGamma parametric fit (Project 1)
- `ggplot2`: Nelson-Aalen and synthesis figures

RESULTS AND INTERPRETATION

Kaplan-Meier Estimator: Overall Survival

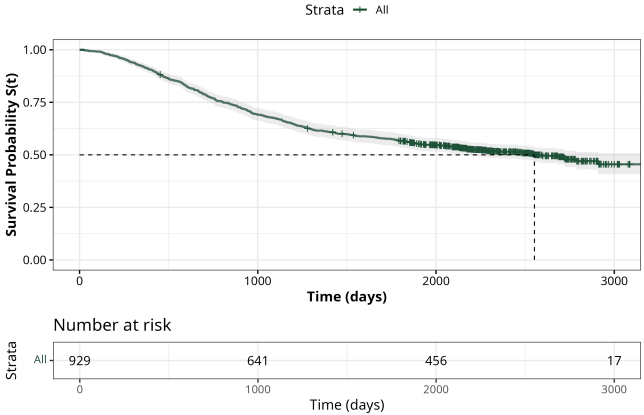

Figure 1: Overall Kaplan-Meier Survival Curve

Table 2: Survival Quantiles for Colon Cancer Patients

Quantile	Days	95% CI
25th	1235	[1106, 1410]
Median	2418	[2152, —]
75th	—	—

- 86% survive at 1 year.
- Median survival ≈ 2 418 days (≈6.6 years), consistent with the GenGamma mean of 2 130 days from Project 1.

Kaplan-Meier Estimator: Stratified by Group

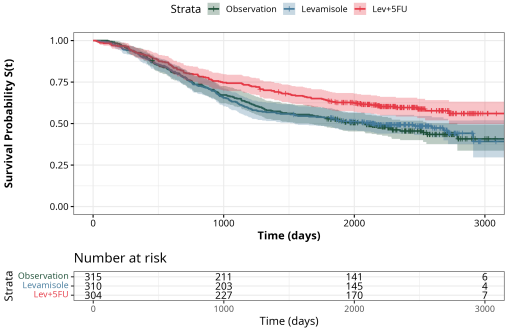

Figure 2: Kaplan-Meier Curve Stratified by treatment

- Lev+5FU separates clearly from Obs and Lev after ∼500 days, indicating a sustained long-term benefit.

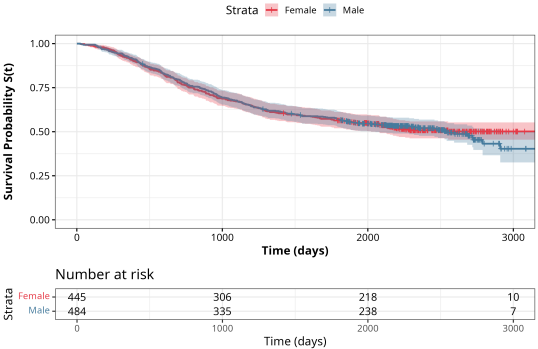

Figure 3: Kaplan-Meier Curve Stratified by Sex

- The female and male curves overlap throughout follow-up, suggesting no meaningful sex effect on survival.

Nelson-Aalen Estimator: Cumulative Hazard & Hazard Rate

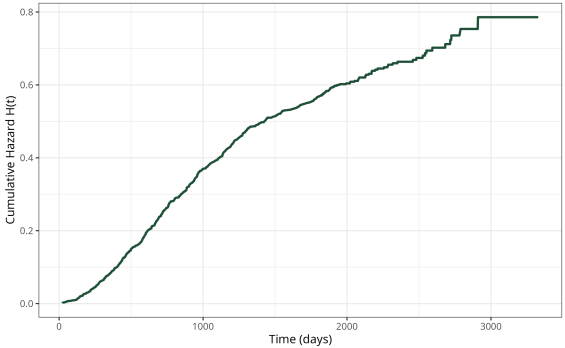

Figure 4: Nelson Aalen Cumulative Hazard

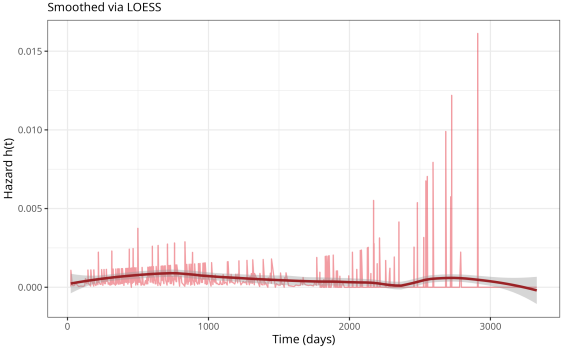

Figure 5: Nelson Aalen (Non-Parametric) Hazard

- $H(t)$ is **concave**, it rises quickly at first then flattens, indicating decelerating cumulative risk.

- The smoothed hazard $h(t)$ is **non-monotone**: it rises to a peak near day 300 then gradually declines. This matches the Generalised Gamma model from the 1st presentation ($Q = -0.481 < 0$), confirming a non-monotone hazard shape.

Log-Rank & Wilcoxon Tests – Hypothesis Testing

Table 3: Log-Rank Vs Wilcoxon Tests for Treatment and Sex.

Comparison	Test	χ^2	p	Decision
Treatment (rx)	Log-Rank	16.79	<0.05	Reject H_0
Treatment (rx)	Wilcoxon	14.45	<0.05	Reject H_0
Sex	Log-Rank	0.40	>0.05	Fail to reject
Sex	Wilcoxon	0.38	>0.05	Fail to reject

Log-Rank vs Wilcoxon

The Log-Rank test assigns equal weight to all event times and is optimal under proportional hazards. The Wilcoxon (Breslow) test weights early events more heavily. Both tests agree here, indicating the treatment effect is consistent across early and late follow-up.

Key findings

- **Treatment (rx):** Both tests reject H_0. Survival differs significantly across treatment groups. Lev+5FU has the best prognosis.
- **Sex:** Both tests fail to reject H_0. No significant survival difference between male and female patients.
- Agreement between both tests suggests the treatment effect is stable over time, not concentrated at early or late failure times.

Cox Proportional Hazards: Univariable Models

For each covariate, a univariable Cox model was fitted using $h(t \mid x) = h_0(t)\, e^{\beta x}$. The table reports the estimated hazard ratio (HR), 95% CI, and significance.

Table 4: Univariable Cox Model with Hazard Ratios

Covariate	Level	HR	Lower	Upper	p-value	Sig
Age (per year)	—	1.005	0.998	1.012	0.1490	
Sex	Male	0.913	0.754	1.106	0.3620	
Treatment	Lev	0.978	0.796	1.202	0.8290	
Treatment	Lev+5FU	0.656	0.530	0.813	<0.001	***
Nodes (per node)	—	1.079	1.066	1.093	<0.001	***
Differentiation	Moderate	1.261	0.893	1.781	0.1840	
Differentiation	Poor	1.638	1.124	2.387	0.0098	**

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

- Lev+5FU, nodes, and poor tumour differentiation are significant univariable predictors. Age, sex, and Lev alone are not significant.

Cox Proportional Hazards: Multivariable Model

The multivariable model includes: age, sex, rx, nodes, and differentiation simultaneously.

Table 5: Multivariable Cox Model with Hazard Ratios and Confidence Intervals

Covariate	HR	95% CI	p-value	Sig
Age	1.004	[0.996, 1.011]	0.3250	
Sex (Male)	0.885	[0.729, 1.074]	0.2190	
Lev	0.973	[0.793, 1.195]	0.7970	
Lev+5FU	0.638	[0.516, 0.789]	<0.001	***
Nodes	1.077	[1.063, 1.091]	<0.001	***
Diff. Moderate	1.214	[0.859, 1.716]	0.2700	
Diff. Poor	1.592	[1.090, 2.325]	0.0156	**

*** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

Interpretation

- **Lev+5FU:** HR = 0.638, a 36% reduction in hazard vs. observation, adjusting for all other covariates.
- **Nodes:** Each additional positive lymph node raises hazard by 7.7%.
- **Poor differentiation:** 59% higher hazard vs. well-differentiated tumours.
- Age and sex remain non-significant after adjustment, consistent with univariable results.

PH Assumption: Schoenfeld Residuals (Diagnostics & Refinement)

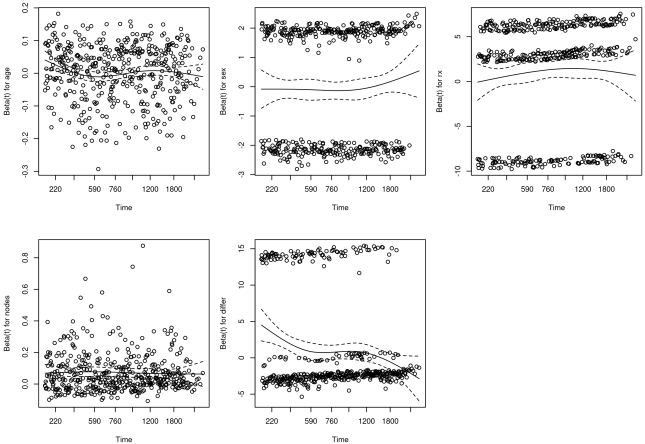

Figure 6: Schoenfeld Residuals Plots

The Schoenfeld residual test formally evaluates H_0: the scaled residuals are uncorrelated with time, i.e. the covariate effect is constant.

Covariate	p	PH?
Age	>0.05	✓
Sex	>0.05	✓
rx	>0.05	✓
Nodes	>0.05	✓
Differ	>0.05	✓
Global	>0.05	✓

■ All $p > 0.05$. The PH assumption holds globally and for each covariate. No stratification or time-varying coefficients are required.

Linearity Check & Influential Observations (Diagnostics & Refinement)

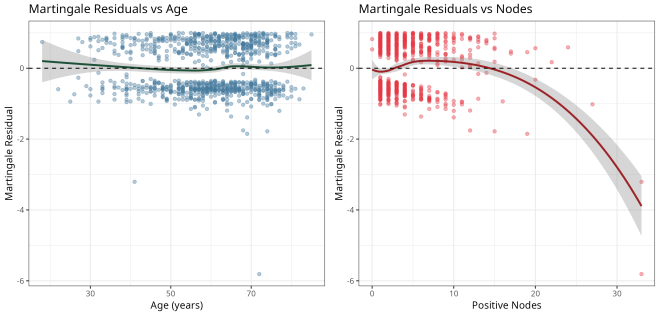

Figure 7: Martingale Residuals against Age and Nodes

■ Martingale residuals plotted against continuous covariates. The LOESS (Locally Estimated Scatterplot Smoothing) trend for `nodes` is non-linear: a steep initial rise that flattens at higher node counts. This suggests the linearity assumption in the log-hazard is violated for nodes.

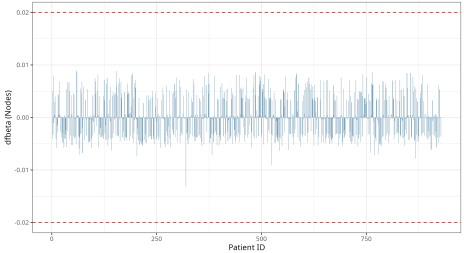

Figure 8: Influential Observations

– **dfbeta** values for `nodes`.

■ All observations fall within ± 0.02, indicating no single patient is distorting the results. The results are stable.

Refined Cox Model: Penalised Spline for Nodes

Figure 9: Non-Linear Effect of Nodes (Penalised Spline)

■ The penalised spline (`pspline(nodes, df=4)`) reveals a **non-linear** log-hazard relationship: risk rises steeply from 0-5 positive nodes, then continues to increase more gradually. The curve is truncated at the 95th percentile to avoid unreliable estimates in the sparse tail.

Table 6: Refined Model

Covariate	HR	95% CI	p-value	Sig
Age	1.004	[0.997, 1.012]	0.3190	
Sex (Male)	0.880	[0.725, 1.069]	0.2060	
Lev	0.970	[0.790, 1.190]	0.7650	
Lev+5FU	0.634	[0.512, 0.784]	<0.001	***
pspline(nodes)	non-linear		<0.001	***
Diff. Moderate	1.212	[0.858, 1.713]	0.2750	
Diff. Poor	1.584	[1.085, 2.314]	0.0170	**

■ The main conclusions are unchanged. The spline improves model fit without altering the clinical interpretation of Lev+5FU, nodes, or poor differentiation.

KM vs. Generalised Gamma (Synthesis)

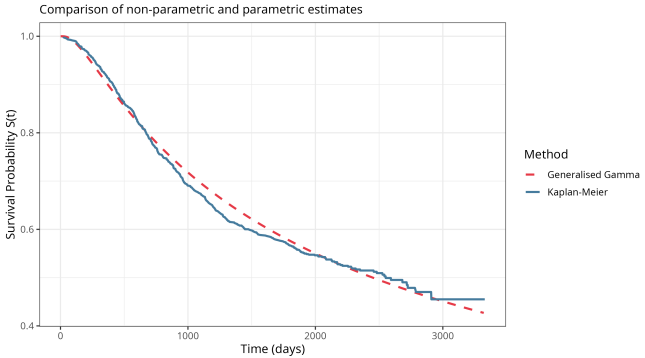

Figure 10: KM Vs Parametric (Generalised Gamma) Survival Curves

■ KM (solid) and Generalised Gamma (dashed) survival curves are nearly identical up to $t \approx 2\,000$ days. They diverge slightly only in the sparse upper tail, confirming excellent parametric fit.

Convergence across methods

1. KM median (≈2 418 days) is close to the GenGamma mean (2 130 days)
2. Nelson-Aalen hazard peak (∼day 300) matches the GenGamma non-monotone shape ($Q = -0.481$) exactly
3. Log-Rank confirms treatment effect; Cox quantifies it: Lev+5FU, HR = 0.638
4. Nodes is the strongest continuous predictor (HR ≈ 1.077 per node)
5. Poor differentiation raises hazard by ∼59% vs. well-differentiated

CONCLUSION

Conclusions

- `Overall survival:` 86% survive 1 year; median $\approx$ 2 418 days. Lev+5FU patients show markedly better long-term survival; sex has no significant effect.
- `Hazard shape:` Non-monotone cumulative hazard. The instantaneous hazard peaks near day 300 then declines, consistent with the GenGamma model ($Q = -0.481$).
- `Group differences:` Treatment significantly affects survival ($p < 0.05$, both tests); sex does not. Log-Rank and Wilcoxon agree, confirming the treatment effect is stable over time.
- `Key predictors:` `Lev+5FU` (HR = 0.638, 36% hazard reduction), `nodes` (HR = 1.077 per node, non-linear), and `poor differentiation` (HR = 1.592).
- `Synthesis:` Non-parametric (KM/NA), parametric (GenGamma), and semi-parametric (Cox) methods consistently identify a high early-risk phase, declining hazard, and a strong protective benefit from Lev+5FU.

Clinical Implications

1. **Prioritise Lev+5FU:** The 36% hazard reduction is robust to covariate adjustment and confirmed by both non-parametric and semi-parametric methods.
2. **Intensify post-operative monitoring in the first 300 days:** This is the period of highest hazard. Closer surveillance could improve early detection of recurrence.
3. **Node count as a risk stratifier:** The non-linear effect means that even moving from 0 to 3 positive nodes substantially elevates risk. Node status should guide adjuvant therapy decisions.
4. **Poor tumour differentiation:** A 59% higher hazard identifies this group as high-priority for aggressive treatment protocols.

References

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Thank You

for your attention

Survival Analysis - Assignment 2

Group 2 — AIMS Rwanda,
March 2026

https://github.com/tchfare/Survival-Analysis-Group2